



The semiconductors and components are bought from suppliers who are two levels below. Which means, the automotive manufacturers are not usually aware of what's happening in the supply chain until it hits them too late. In such a complex scenario, Industry 4.0 can make a major difference by adopting the semiconductor industry. Even with semiconductors being the driving force behind technologies like artificial intelligence (AI) and machine learning (ML), the semiconductor industry in itself has not seen much of Industry 4.0.

To summarise, semiconductor test solution providers are looking to change this in a way where the semiconductor industry itself is benefitted. It might be important to note that Advantest generates the most amount of data in the semiconductor industry across all players. We are using this data to take the industry to the next level in terms of digital transformation.

Would better visibility in the supply chain be of help?

Yes, and that is one of the essential things being worked on today. As an industry, we are trying to build solutions over the cloud that will allow any manufacturer to calculate what's happening in the supply chain well ahead of time. The solutions will tell manufacturers about problems that can arise in the future, giving them enough time to react and come up with a resolution.

The cloud is one function that we are most excited about. When talking about increasing the yield through a fab, it would take at least a year. But when it comes to establishing a new fab, it could take around two to three years of time. However, with Industry 4.0 in the picture, we can guarantee improved yield starting today. Yet these are just some functions of the Industry 4.0 value chain.

How Industry 4.0 can complement semiconductor industry

1. Elements of Industry 4.0 can be used to validate semiconductor design from the very first step.
2. Design engineers do not need to wait for 2.5 months to gauge the effectiveness of the die they are working on.
3. A semiconductor travels more than 48,000 kilometres (30,000 miles) before reaching the consumer. Industry 4.0 can be used to collect data from all the points of distance it covers.
4. There is better visibility of the supply chain, both for the creators as well as users of the semiconductors.

Is it expensive?

Any transformation requires investment. The focus, however, is mostly on the return on investment (ROI), and early use cases have already proven an impressive ROI. Adding to it, cases that come online would only increase ROI further. The best way to test waters for semiconductor companies is by signing up for subscription models instead of making all investments before the value is understood. If the results are satisfactory, such companies can fine tune their investments.

Will it reduce jobs?

As a matter of fact, Industry 4.0 creates more jobs. The need of the hour for professionals working in the semiconductor industry is to up-skill and re-skill themselves. Almost every semiconductor industry needs data scientists, and this explains the magnitude of jobs that can evolve.

What are the other possible areas where Industry 4.0 can be used in the semiconductor industry?

The complexity of the semiconductor manufacturing process requires high levels of visibility, analytics, and awareness. Analytics in the past was mostly manual. Let me explain this through an analogy.

Just like how a multiple-use residential/commercial skyscraper consists of many layers—structures with interdependencies—similarly, a semiconductor device process is like a skyscraper that has many floors or processing layers, as we call them in the industry. It takes 2 to 2.5 months to put these layers onto a device for a typical semiconductor company.

A lot can go wrong in this process. The smallest mistakes can affect the time of the semiconductor to market by weeks, or in some cases months. This in turn affects the overall image of a semiconductor company. What makes the process more challenging is that different layers of the device are being worked upon in different parts of the world.

A typical semiconductor travels no less than 48,000 kilometres.

Now all the places a semiconductor goes, tremendous amounts of data is generated. Sometimes it is collected manually as well. The data acquired is in multiple languages, and one of the biggest challenges that we as an industry face is to bring this data under one machine in one language to gauge what's going right and what's not.

Industry 4.0 provides the analytics of this data and improves the yields that directly improve the cost. It can also help in answering the questions that are related to the tests of semiconductors. Issues of over testing or under testing can be answered with the help of Industry 4.0. Though there are many more advantages that it offers, the target of most companies is to decrease the time to market for the semiconductors.

What all stages of semiconductor manufacturing can benefit from Industry 4.0?

A term often referred to in the semiconductor industry is the